

SCE-003 — IN-SITU KNOWLEDGE GROWTH

Why Knowledge May Grow in Place Rather Than Being Rewritten

Structural Cognitive Emergence (SCE)

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Repository: <https://github.com/sizhet/Structural-Cognitive-Emergence-SCE>

Introduction

One of the most common assumptions about learning is that new knowledge replaces old knowledge.

This assumption appears natural.

When we learn something new, it often feels as though the brain updates itself.

Many modern AI systems operate in a similar way.

Training adjusts parameters.

New information modifies existing representations.

The system gradually changes over time.

However, biological cognition may operate differently.

This document explores a different possibility:

Knowledge may primarily grow in place rather than being repeatedly rewritten.

If true, this simple idea may help explain:

- prenatal learning,
- infant development,
- childhood concept formation,
- scientific expertise,
- lifelong learning,
- and even the emergence of curiosity itself.

The Tree Analogy

Consider a tree.

A mature tree is not rebuilt every year.

The tree does not erase its previous structure and start again.

Instead:

Root

↓

Branch

↓

Branch

↓

Branch

↓

Leaf

new structures emerge around existing structures.

Growth is additive.

Growth is local.

Growth is cumulative.

Most importantly:

The tree grows in place.

SCE proposes that biological knowledge may behave similarly.

Learning Before Birth

One of the most intriguing observations in developmental psychology is that learning appears to begin before birth.

Numerous studies suggest that newborns can recognize:

- maternal voices,
- familiar rhythms,
- recurring sound patterns,
- and other prenatal sensory experiences.

The details remain debated.

However, the general observation is difficult to ignore:

Some form of memory appears to exist before formal learning begins.

Why?

The traditional explanation is often:

Information Storage

SCE proposes a complementary interpretation:

Structure Formation

The fetus may not be acquiring explicit knowledge.

Instead, it may be constructing initial structural scaffolding.

Pattern Before Meaning

A fetus does not understand language.

Yet recurring sounds may still matter.

Why?

Because pattern recognition can exist before semantic understanding.

Repeated exposure to:

Voice A

Voice B

Rhythm A

Rhythm B

creates differential structures.

The brain begins organizing:

Similar

Different

Expected

Unexpected

long before concepts or language emerge.

Structural growth may therefore precede meaning.

The Infant Phase

After birth, the process accelerates dramatically.

The infant encounters:

Faces

Voices

Movements

Objects

Textures

Temperatures

The brain does not merely collect observations.

The brain continuously expands existing structures.

Every new experience becomes attached to previous structures whenever possible.

Knowledge therefore grows around existing anchors.

Rather than:

Erase

Rewrite

Replace

the dominant process may be:

Attach

Expand

Differentiate

Concept Formation Revisited

The Cat Problem provides an excellent example.

Suppose a child encounters:

Cat A

A small structure forms.

Later:

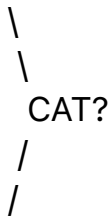
Cat B

arrives.

The first structure is not discarded.

Instead:

Cat A



Cat B

begins to emerge.

When:

Cat C

appears,

the structure becomes stronger.

The concept emerges not because previous observations are rewritten.

The concept emerges because previous observations are preserved and expanded.

Knowledge as Structural Accumulation

This leads to a different interpretation of learning.

Traditional view:

Knowledge

+

New Information

↓

Updated Knowledge

SCE view:

Knowledge

↓

Structural Expansion

↓

Larger Knowledge Structure

The difference appears subtle.
The implications are profound.
The first view emphasizes replacement.
The second view emphasizes growth.

Why Children Learn So Quickly

Children may learn rapidly because much of their cognitive effort is devoted to structure construction rather than structure maintenance.

Their cognitive landscape is expanding rapidly.

Every new observation has many potential attachment points.

This creates an environment where:

Differential Structures

form quickly,

Concepts

emerge frequently,

and

Questions

appear naturally.

Learning becomes growth rather than correction.

Why Experts Learn Differently

Experts do not stop growing.

However, growth changes form.

A scientist with decades of experience already possesses extensive cognitive structures.

New observations rarely create entirely new foundations.

Instead:

Existing Structure

↓

Local Expansion

↓

New Connections

becomes dominant.

This explains why expertise often feels less like memorization and more like integration.

The expert is not replacing knowledge.

The expert is extending a mature cognitive tree.

The Scientific Discovery Process

Scientific breakthroughs often appear sudden.

Yet they usually emerge after years of accumulation.

For example:

Observation

Observation

Observation

Observation

may continue for decades.

The scientist appears stuck.

Then suddenly:

Insight

appears.

SCE interprets this differently.

The insight may not be sudden creation.

The insight may be the visible result of long-term structural growth.

The tree was expanding continuously.

The final branch simply became visible.

In-Situ Growth and Curiosity

An important consequence follows.

If structures grow in place,

then every expansion may expose new boundaries.

Every new boundary may reveal:

Unknown Region

These unknown regions become:

Structural Gaps

The appearance of gaps naturally generates:

Curiosity

Curiosity therefore becomes a byproduct of growth.

The system asks questions because the structure expanded far enough to discover what it does not yet know.

Why Children Ask So Many Questions

This perspective offers a different interpretation of childhood curiosity.

Children do not ask questions merely because they lack knowledge.

Children ask questions because their cognitive structures are expanding rapidly.

Rapid expansion creates:

New Boundaries

New boundaries create:

New Gaps

New gaps create:

New Questions

Question generation therefore becomes evidence of growth rather than ignorance.

Education Reconsidered

This perspective also raises an important educational question.

If knowledge primarily grows in place,
then education may not be the transfer of information.

Instead, education may be:

Structure Cultivation

The most effective teacher may not be the one who provides the most answers.

The most effective teacher may be the one who helps students construct strong structures upon which future growth can occur.

Implications for Artificial Intelligence

Most modern AI systems focus on parameter optimization.

SCE suggests that biological intelligence may emphasize:

Structural Preservation

Structural Expansion

Differential Growth

Gap Discovery

Future AI systems may therefore require mechanisms for maintaining and extending persistent cognitive structures rather than repeatedly reconstructing them from scratch.

The Central Hypothesis

The central hypothesis of this document is:

Biological knowledge may primarily grow through structural expansion around existing cognitive structures.

In this view:

Observation

↓

Attachment

↓

Differential Expansion

↓

Structural Growth

↓

Gap Discovery



Further Growth

Knowledge is not continuously rewritten.

Knowledge grows.

And understanding how knowledge grows may be essential for understanding how concepts, questions, curiosity, and intelligence itself emerge.

Next

SCE-004

Calling Graph Emergence

Why concepts may behave more like expanding calling graphs than isolated labels.